Resilient Normative Alignment for Autonomous Agents

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Resilient Normative Alignment for Autonomous Agents

In this work:

- **Resilient** agents avoid, withstand, recover from, adapt, and evolve to handle uncertainty, change, faults, failure and adversity – gracefully and with minimal disruption
- **Normative alignment** involves compliance with the appropriate, finite set of social, legal, ethical, empathetic and cultural (SLEEC) rules
- **Autonomous agents** are entities that perceive their environment through sensors, make reasoned decisions, and act upon them via actuator
  - focus is on agents with bounded capabilities (finite set of actions/options)
  - extension to foundation-model agents represents future work

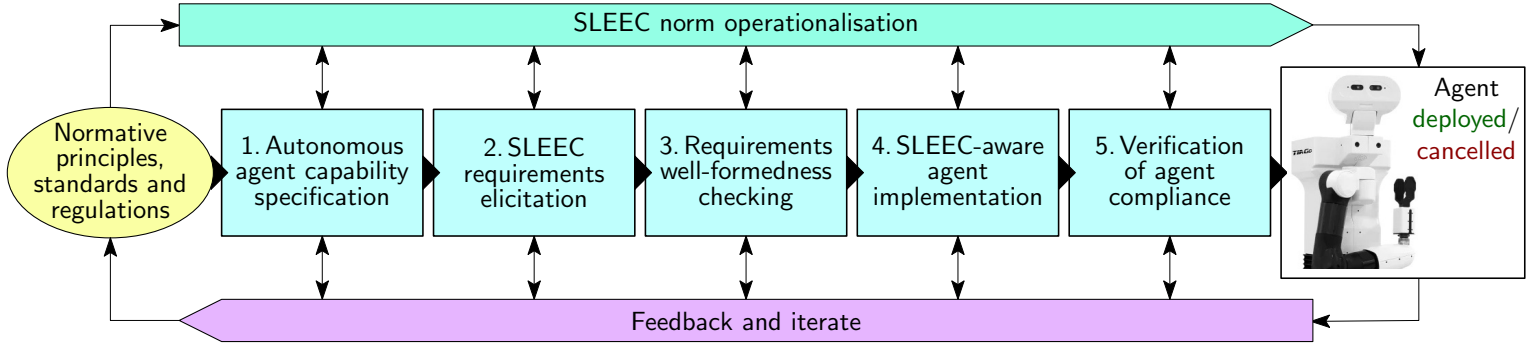
Resilient Normative Alignment for Autonomous Agents

Extension of SLEEC framework presented in yesterday's tutorial

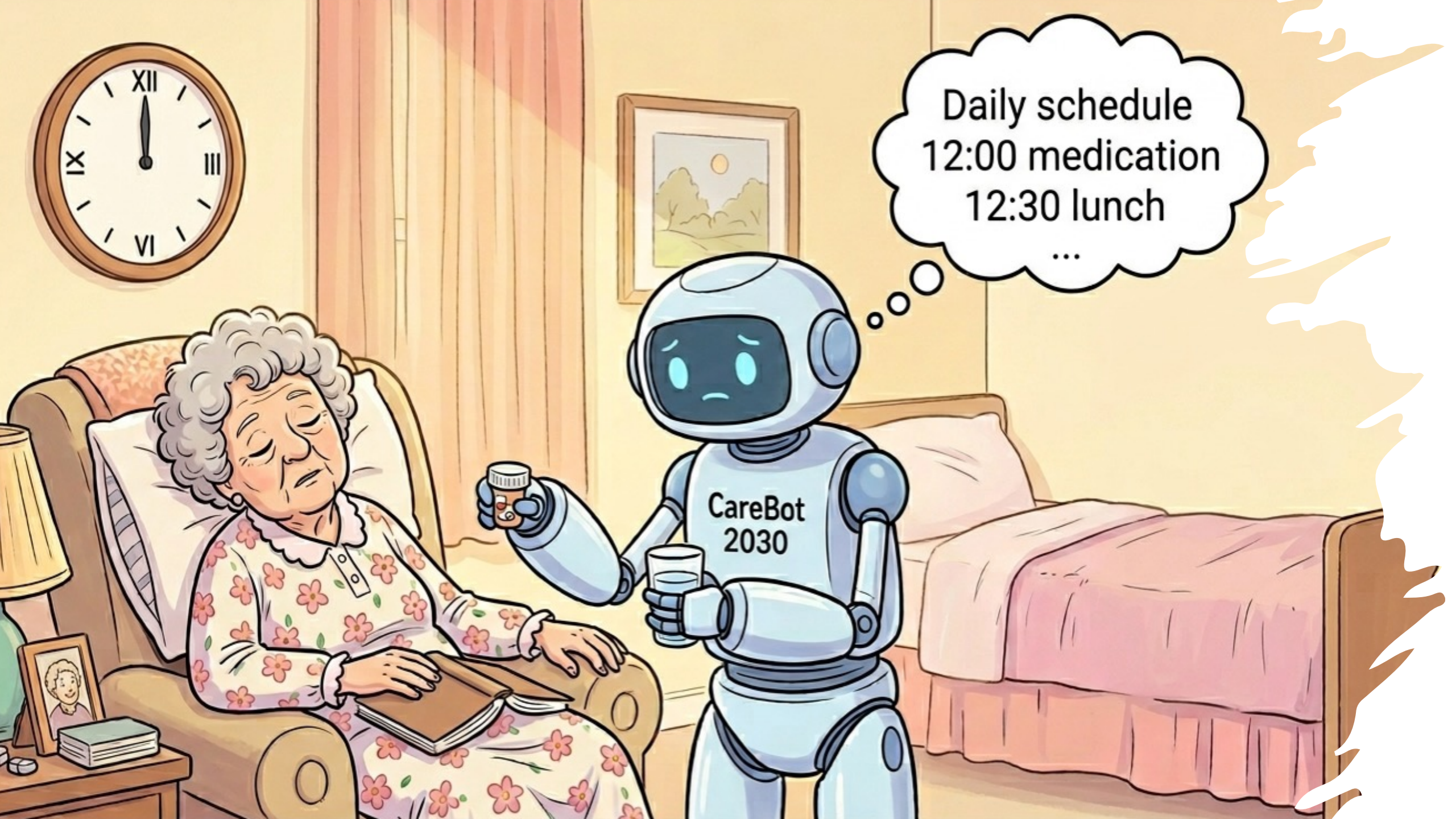
- domain-specific language for defining SLEEC rules

  when UserFallen **then** SupportCalled **within** 30 seconds **unless not** UserAssent

- methods for SLEEC rule elicitation, well-formedness validation, and agent-compliance verification
- tool-supported SLEEC-norm operationalisation methodology



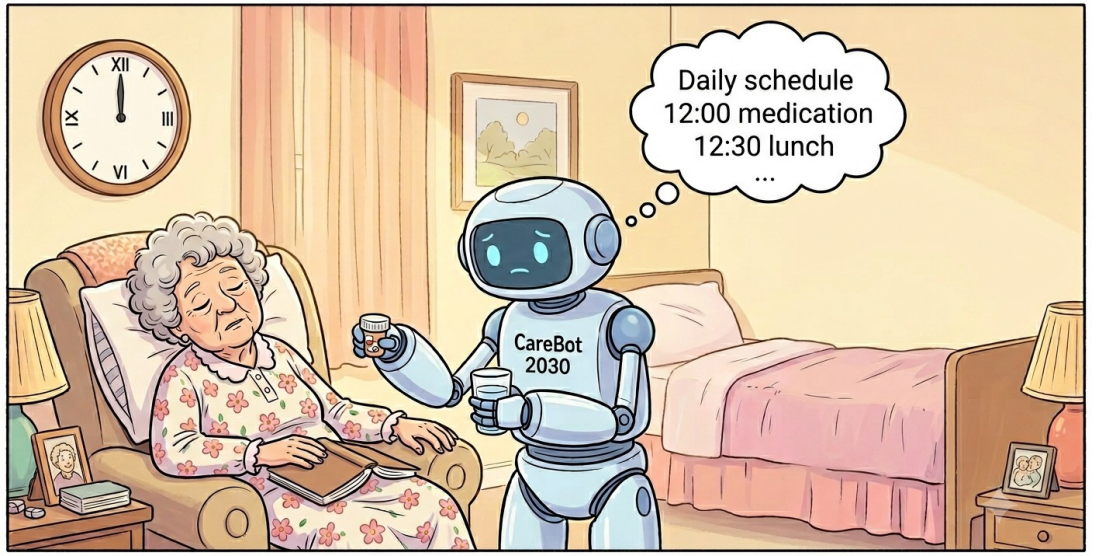
- use cases


Daily schedule
12:00 medication
12:30 lunch
...
CareBot
2030

Two important challenges (of many)

1. While the agent's core functionality is often predefined (give medication, offer lunch), end-user SLEEC requirements differ
2. Formal verification only guarantees an agent's compliance with a SLEEC ruleset under numerous (and often implicit) assumptions, e.g.:
    - agent's sensors/effectors are operational
    - environment conditions (light level, temperature) remain within anticipated bounds
    - users responses are within expected limits

Typically, human operators/carers handle both challenges seamlessly



Two important challenges (of many)

1. While the agent's core functionality is often predefined (give medication, offer lunch), end-user SLEEC requirements differ
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    - agent's sensors/effectors are operational
    - environment conditions (light level, temperature) remain within anticipated bounds
    - users responses are within expected limits

Solutions

1. **SLEEC-ADAPT** – agent workflow adaptation to current user's SLEEC requirements

    A L Partis, Radu Calinescu, Ana Cavalcanti, Sinem Getir Yaman, Alvaro Miyazawa, Ioannis Stefanakos, A MAPE-K Instantiation for Dependable Normative Adaptation in Intelligent Agents, ISSRE 2026.

2. Avoidance of a **SLEEC-FIASCO** through uncertainty management

    Lina Marsso, Isobel Standen, …, Ana Cavalcanti, Radu Calinescu, Marsha Chechik. Dodging a SLEEC-FIASCO: Normative Assurance under Uncertainty. 2026. Under review



SLEEC-ADAPT – agent workflow adaptation to current user's SLEEC requirements

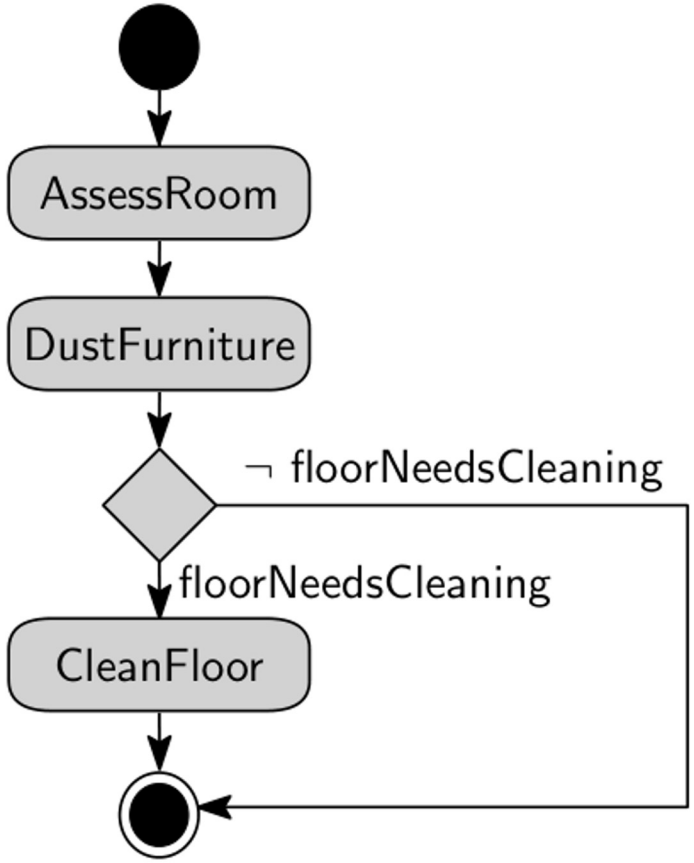
(SLEEC-ADAPT slides provided by Annabelle Partis)

Motivating Example: Service Robot in a Hospital

The robot must dynamically adapt its behaviour to patient SLEEC requirements

Robot capabilities support the tasks

Patient SLEEC rules must map to pre-existing capabilities



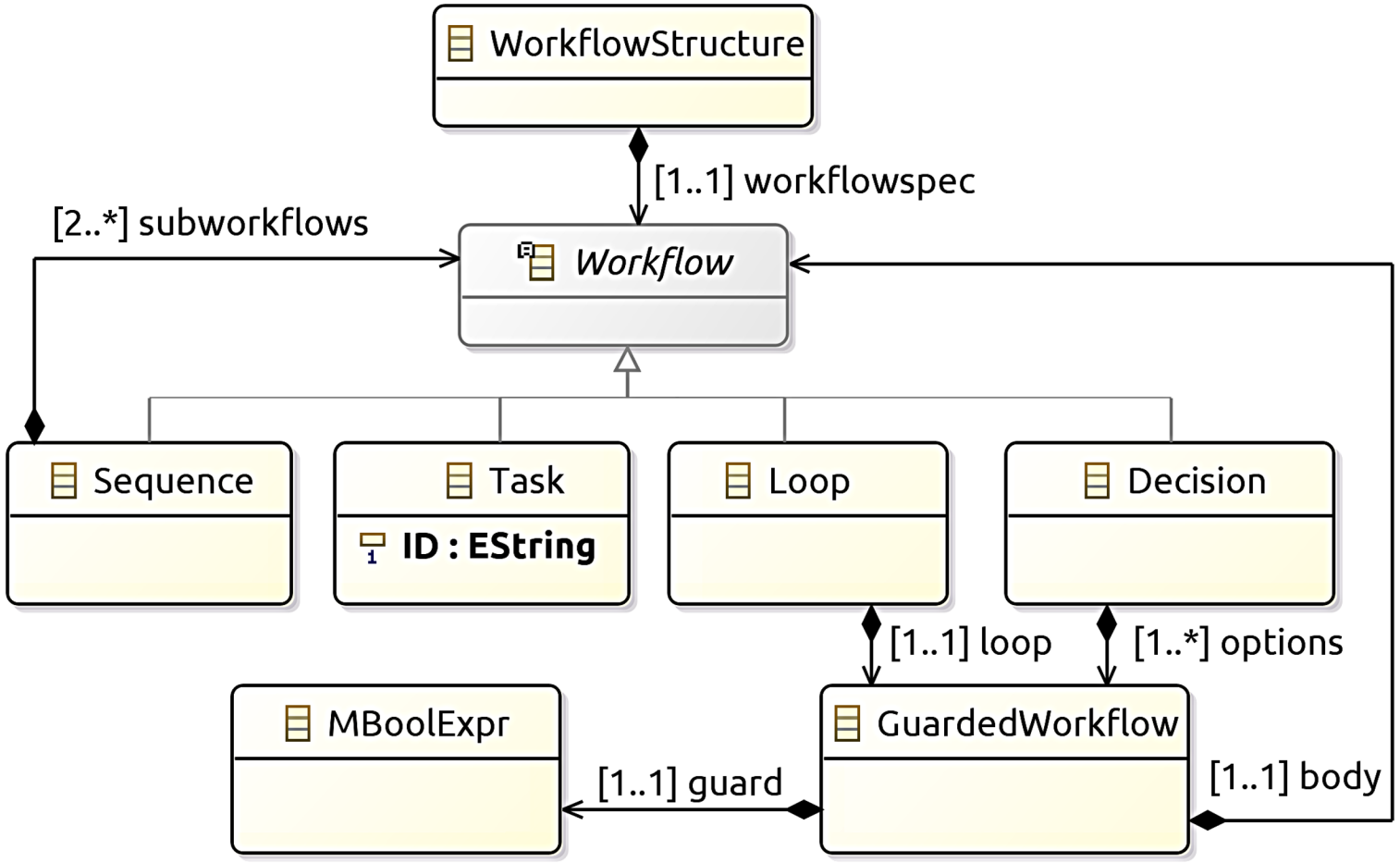
Preliminaries - SLEEC Language

ID	Requirement	SLEEC Type
R1a	Before the room assessment begins, the robot shall notify the patient	Social
R1b	When the patient is resting, the robot shall switch to a silent cleaning mode before cleaning the floor.	Empathetic

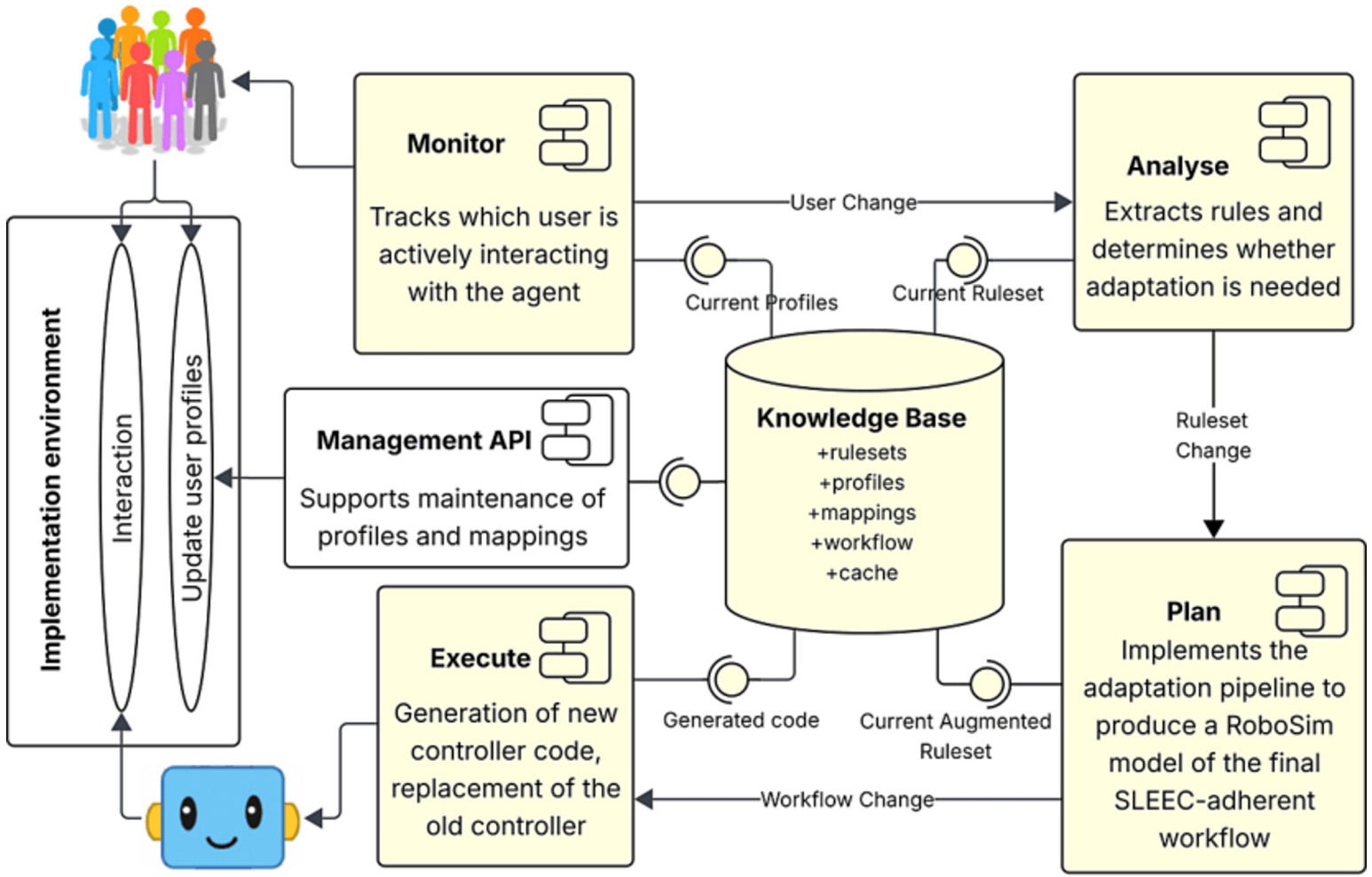


R1a when AssessRoomStart then NotifyPatient
R1b when CleanFloorStart and PersonResting then UseSilentFloorCleaning

Preliminaries - Workflow language



The MAPE-K Architecture



Defining SLEEC rulesets



→ Default SLEEC ruleset



→ SLEEC ruleset for partially sighted user



→ Anxious user SLEEC ruleset

Determining user type

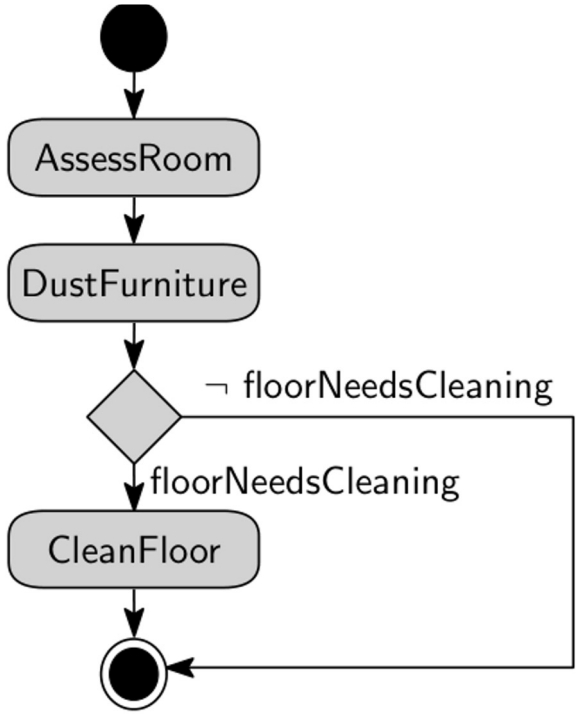


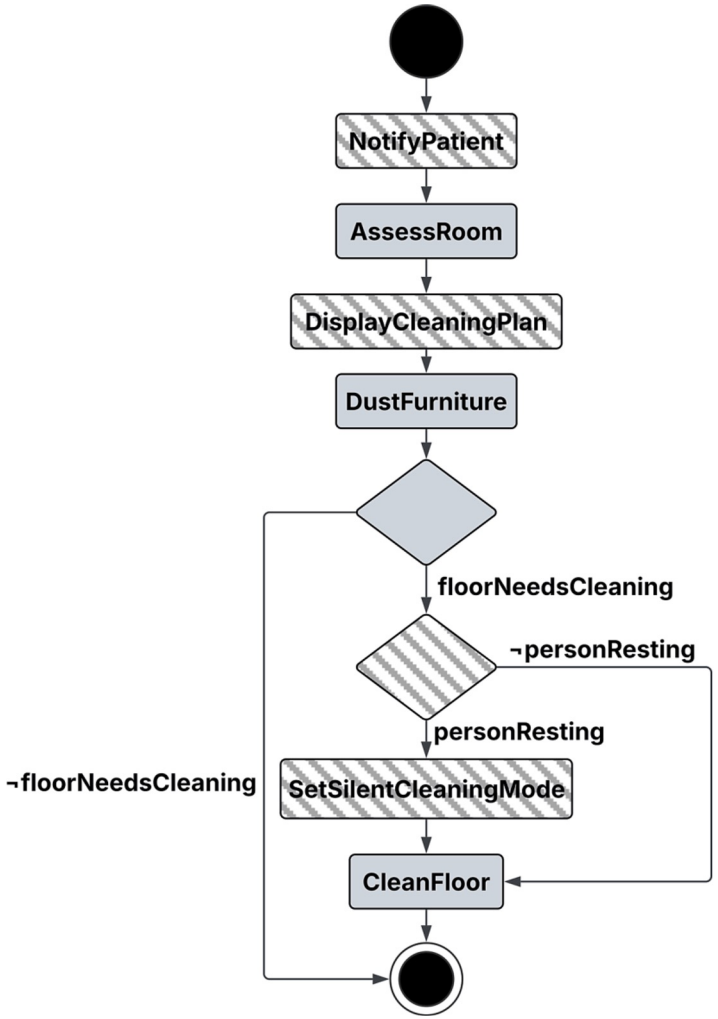
Please specify if you:

☑ are a first-time user ☐ have anxiety . . .

Planning Component

Rule1 when AssessRoomEnd then DisplayCleaningPlan
Rule2 when AssessRoomStart then NotifyPatient
Rule3 when CleanFloorStart and personResting
then SilentFloorCleaning



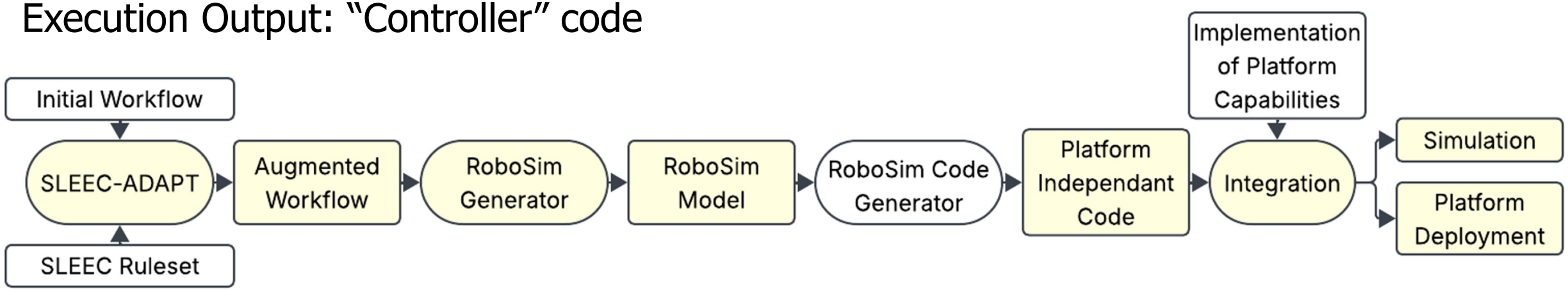


Planning and Execution Components

Planning Inputs: SLEEC ruleset and workflow

Planning Output/Execution Input: SLEEC-Adherent RoboSim specification

Execution Output: "Controller" code



Evaluation: Research Questions

R1 (Correctness): Do all components of our MAPE-K instantiation operate as intended?

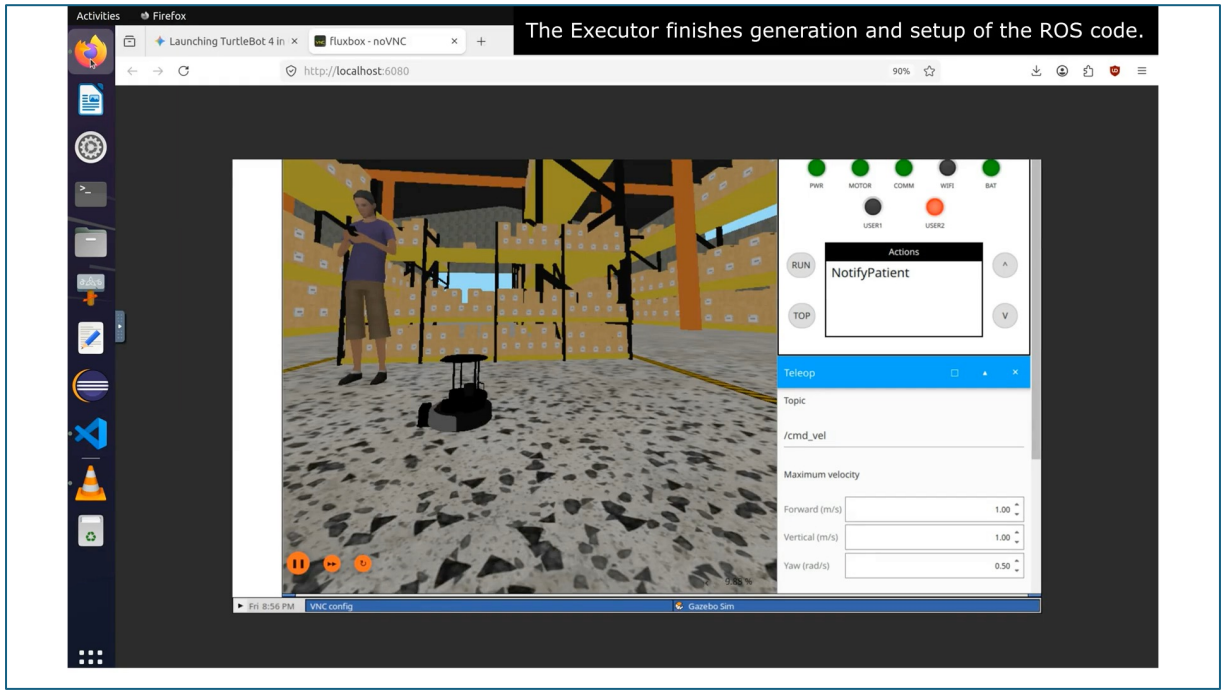
R2 (Breadth of Applicability): Is our proposed architecture applicable to different types of intelligent agents from different application domains?

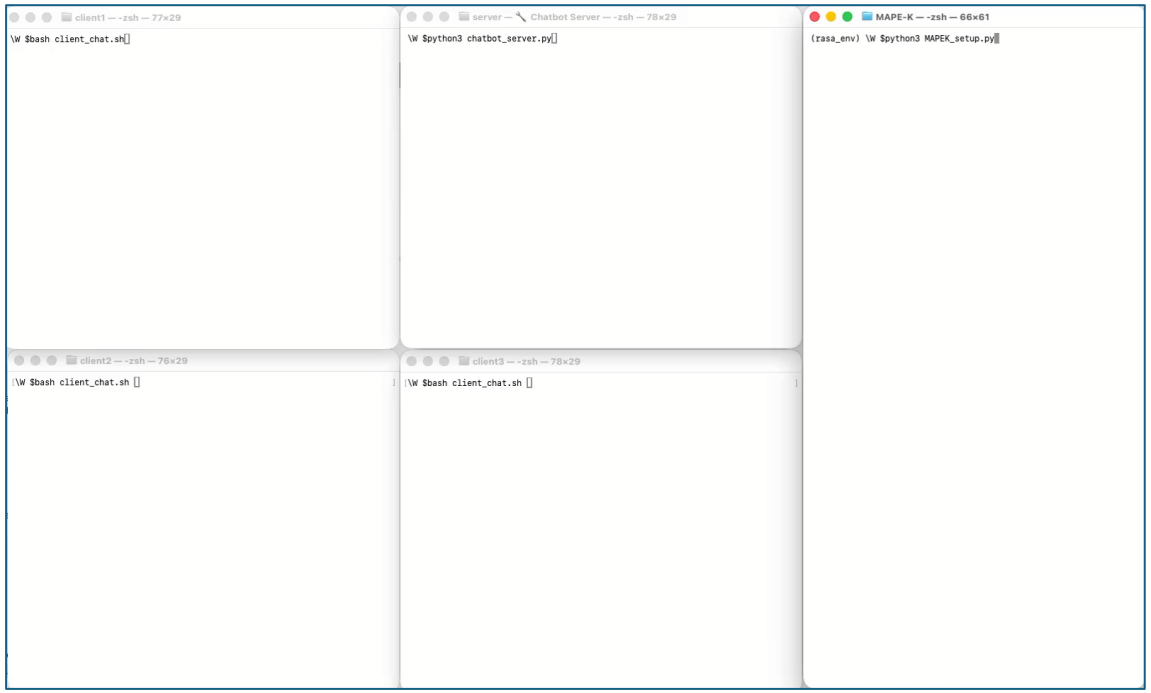
R3 (Scalability and Overhead): Can workflows of realistic size can be augmented and executed with low latency?

Correctness

- Formal proofs (algorithm)
- Branch testing (implementation)
- Trace testing (controller synthesis)

Breadth of Applicability – Service Robot, Customer Service Chatbot



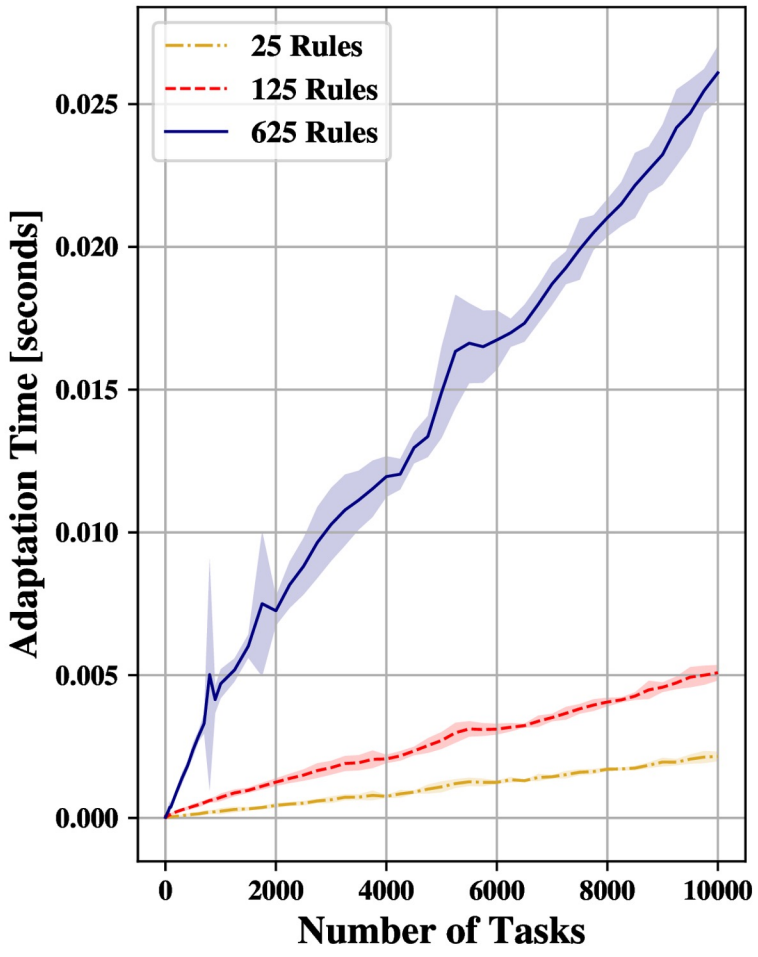


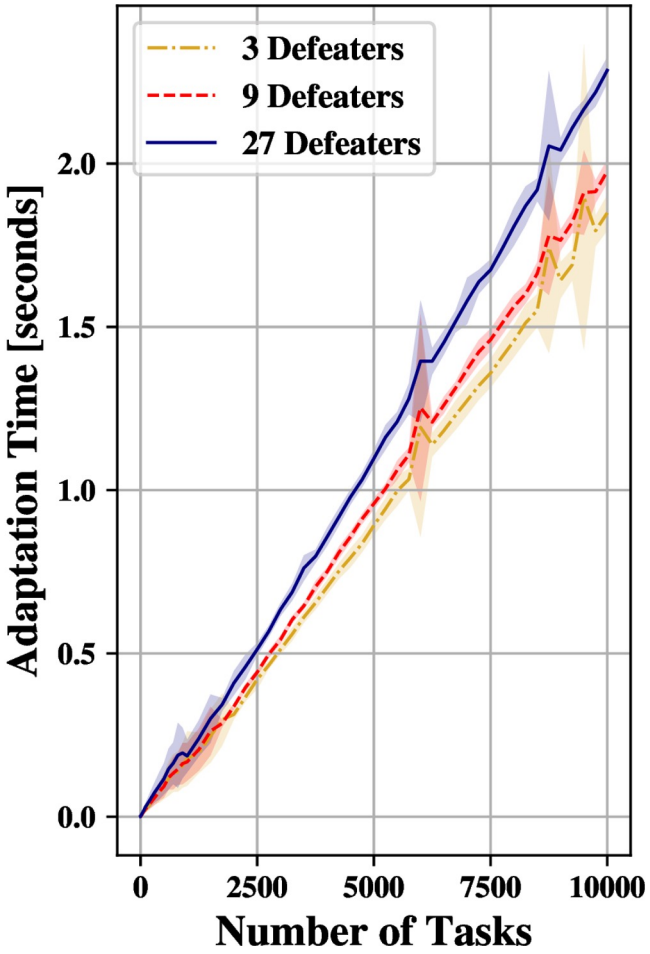
Scalability and Overheads

- Formal proof of adaptation algorithm's linear time complexity

- Backed up with experimentation

The execution phase takes longest - optimisation must be explored.





Dodging a SLEEC-FIASCO: Normative Assurance under Uncertainty



(SLEEC-FIASCO slides provided by Lina Marsso)

Identifying SLEEC disruptions

- Self-adaptive solutions interacting with humans must adapt and evolve while adhering to SLEEC norms
- These solutions often encounter disruptions, such as environmental changes or user-specific challenges

Disabilities being unsupported



Limited social awareness



User impersonating another



- **SLEEC relevant disruption**: An action that prevents the execution of an autonomous system from complying with SLEEC rules as usual or expected.
- **Identifying SLEEC-relevant disruptions requires collaboration with multidisciplinary non-risk-analysis experts** *(absence of regulations and standards)*

Our goal is to develop:
A tool-supported methodology *guiding and supporting multidisciplinary stakeholders in identifying SLEEC-relevant disruptions and reviewing possible mitigation strategies*

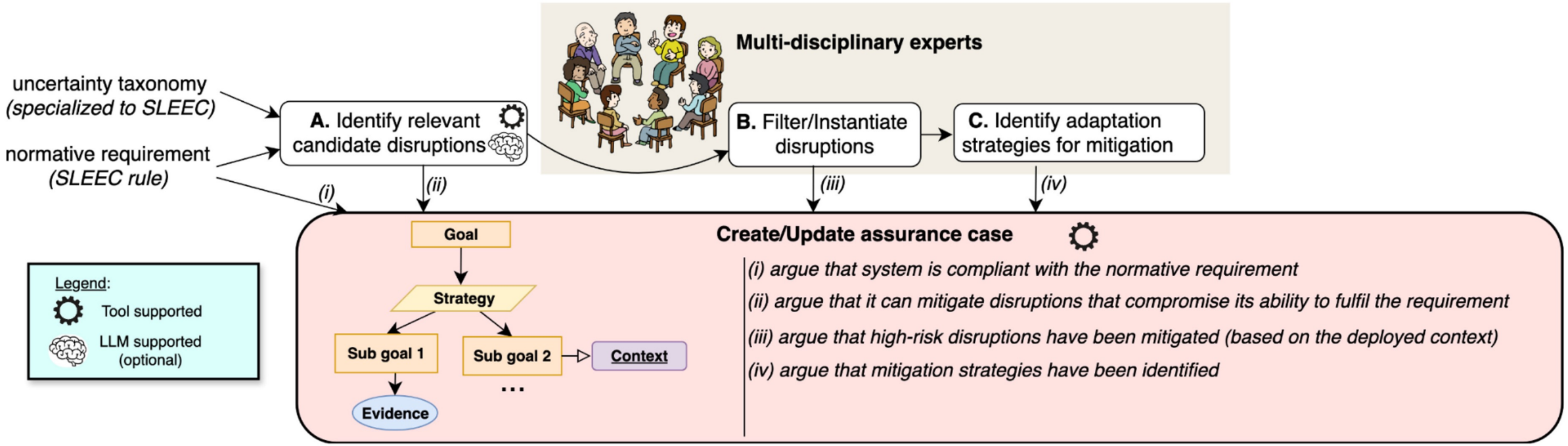
Overview of our approach

1. An uncertainty taxonomy defining disruptions relevant to addressing normative concerns in robotic systems
2. Assurance case patterns for arguing compliance with SLEEC requirements through disruption mitigation
3. An end-to-end, tool-supported methodology that leverages the uncertainty taxonomy and exploits the assurance case patterns to identify SLEEC-relevant disruptions and corresponding mitigation strategies

Dodging a SLEEC-FIASCO (overview)

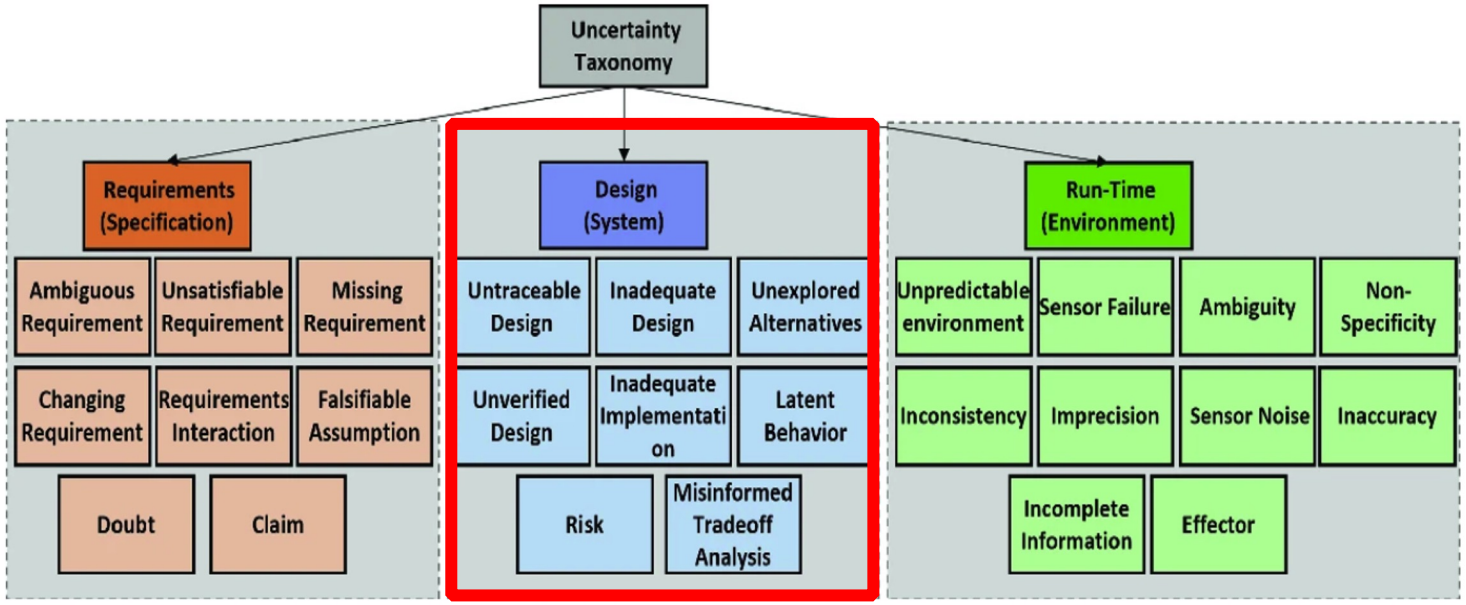
Inputs: a normative requirement expressed as a SLEEC rule and uncertainty taxonomy

Outputs: SLEEC-relevant disruptions with their corresponding mitigations all brought together in an assurance case



Existing taxonomies of uncertainty

Efforts have been made to explore uncertainty in more detail and devise taxonomies



A Taxonomy of Uncertainty for Dynamically Adaptive Systems - Andres J. Ramirez, Adam C. Jensen, and Betty H.C. Cheng

Location
- Context uncertainty
- Model structural uncertainty
- Input parameters uncertainty

Level
- Lack of uncertainty
- Lack of knowledge. Lack of knowledge and lack of awareness.
- Lack of process to find out the lack of awareness.
- Meta uncertainty.

Nature
- Epistemic uncertainty
- Aleatory uncertainty

Uncertainties in the modeling of self-adaptive systems: A taxonomy and an example of availability evaluation - Diego Perez-Palacin and Raffaela Mirandola

But this is still vague for multidisciplinary stakeholders, e.g. "unpredictable environment" could refer to the environment acting as an obstacle to the system's capabilities, or as introducing ambiguity in interpretation.

ADVICE Taxonomy 1/2

ADVICE:
absent knowledge violating SLEEC requirements

Four key dimensions: **Extent, Location**, Nature, Level.

- **Extent** - Fully failing or Partially failing

Full failure: complete inability to capture any intent to initiate the event
Partial failure: some indication of intent, but the event was either not completed (e.g., due to an obstacle or interruption), or the system believed that it succeeded, even though the event was not actually completed.

- **Location** - Environment or Sensor/Actuator

E1. Unpredictable environment acting as obstacle
E2. Unpredictable environment causing ambiguity
E3. Unpredictable environment totally unknown

SA1. Sensor/Actuator Failure
SA2. Sensor/Actuator Imprecision
SA3. Sensor/Actuator Noise
SA4. Sensor/Actuator Inaccuracy

E.g. **obstacle (E1)**:
Disinfection robot cannot start because obstacle on the floor blocks its path

E.g. **actuator inaccuracy (SA4)**:
Mis-calibrated spraying actuator releases an overdose of disinfectant

ADVICE Taxonomy 2/2

ADVICE:
absent knowledge violating SLEEC requirements

Four key dimensions: Extent, Location, **Nature, Level.**

- ★ **Nature** - Epistemic or Aleatory uncertainty

Epistemic: disruption caused by incomplete or outdated knowledge
Aleatory: disruption caused by inherent randomness

- ★ **Level** -

L0 Lack of uncertainty
i.e., no missing knowledge (epistemic)
i.e., deterministic (aleatory)

L1 Known unknown
i.e., awareness of missing knowledge (epistemic)
i.e., awareness of aleatory uncertainty (aleatory)

L2 Unknown unknown
i.e., lack of awareness of missing knowledge (epistemic)
i.e., lack of awareness of aleatory uncertainty (aleatory)

L3 Lack of process
i.e., no process to gain awareness of missing knowledge

E.g. **L0 aleatory**:
No knowledge is missing because of an unpredictable environment

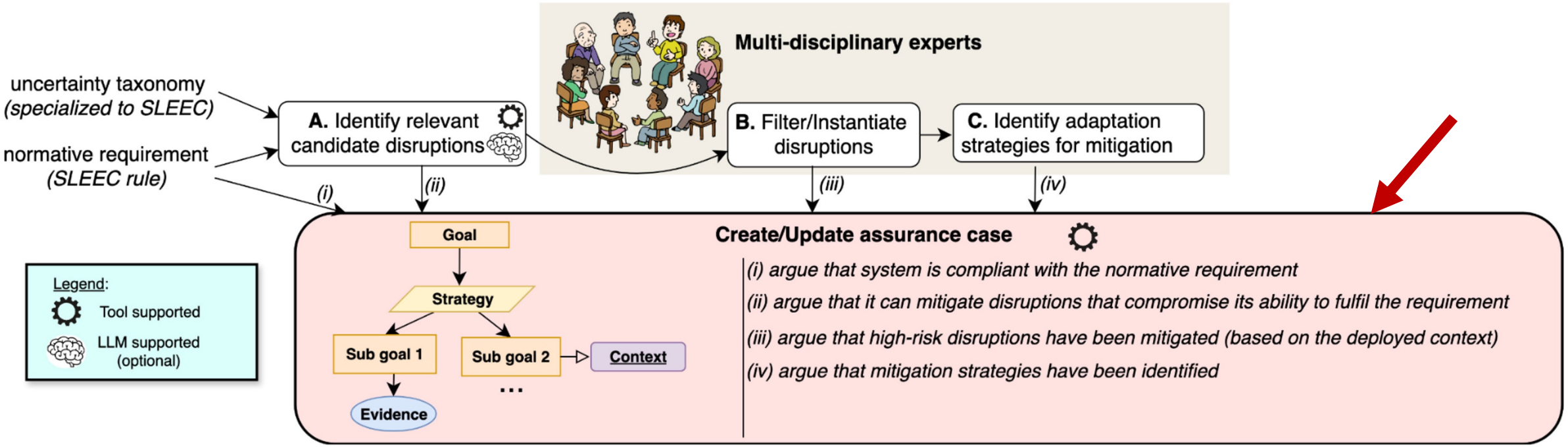
E.g. **L2 epistemic**:
Missing knowledge about whether the actuator fully/partially failed

E.g. **L3 epistemic**:
Unachievable awareness of missing knowledge about sensor failure

Dodging a SLEEC-FIASCO (overview)

Inputs: a normative requirement expressed as a SLEEC rule and uncertainty taxonomy

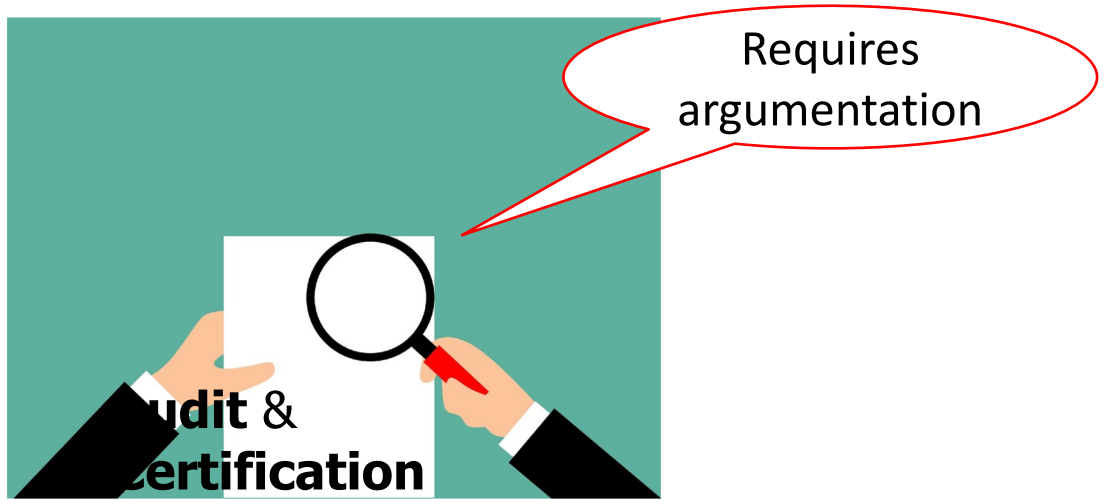
Outputs: SLEEC-relevant disruptions with their corresponding mitigations all brought together in an assurance case

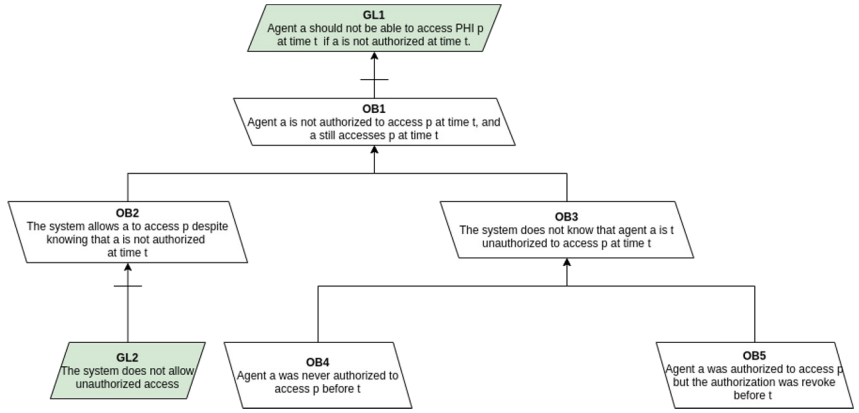


Assurance case (GSN notations)

The assurance case argues that the system is compliant with a given safety and privacy goal by:

- Satisfying the goal directly, when possible
- Mitigating obstacles, when direct satisfaction is not possible
- Operating under explicit assumptions about the environment
- Providing evidence as supporting clues

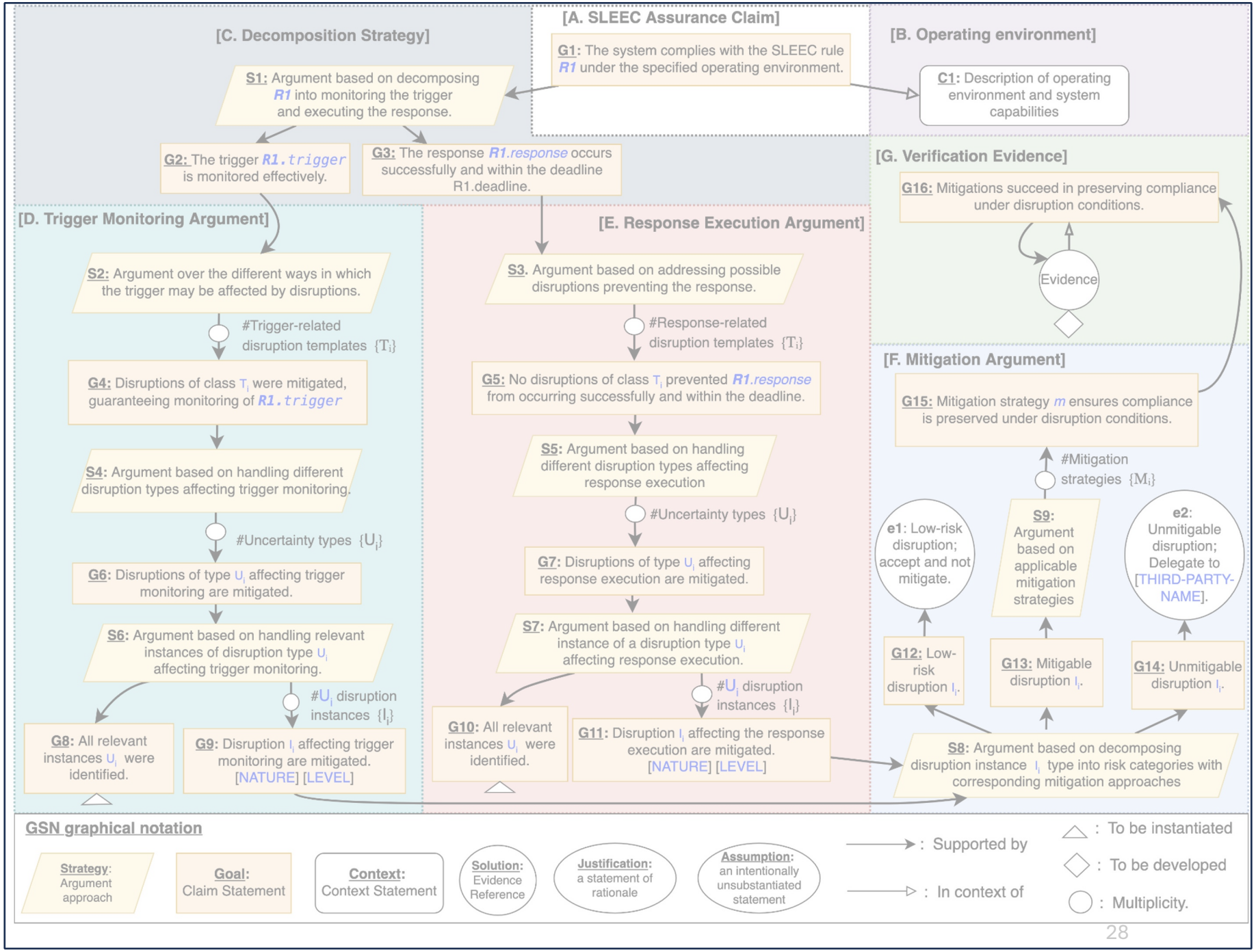




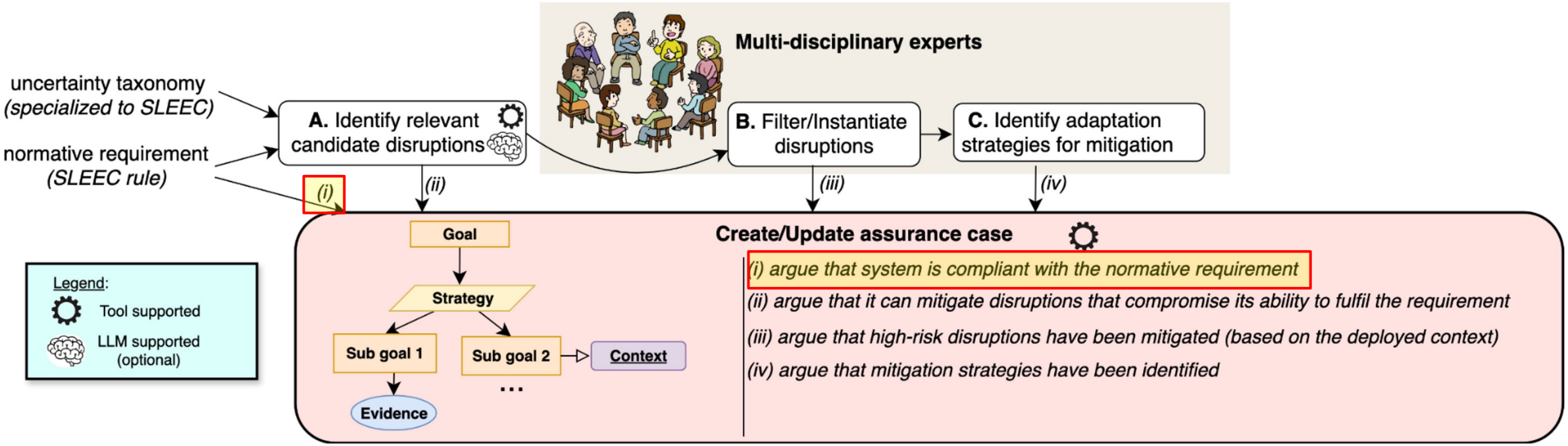
Structured arguments backed up by evidence to justify that a system is compliant in its intended operating environment

Assurance case patterns

Decomposing compliance arguments around trigger, and response failures and their mitigations



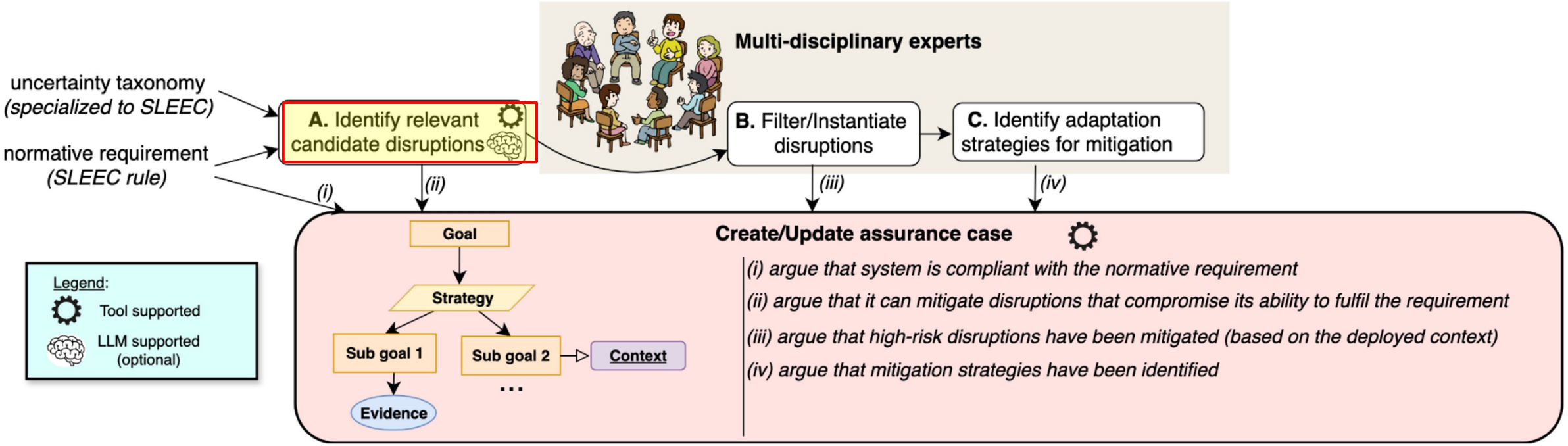
SLEEC-FIASCO: create AC



Create an assurance case with the SLEEC requirement as the main goal of the argument

Goal 1:
The system comply with the SLEEC rule: when PrepareExamination **then** EnsurePrivateSpace **within** 5 **minutes**

SLEEC-FIASCO: identify relevant disruptions



Identify relevant disruptions classes

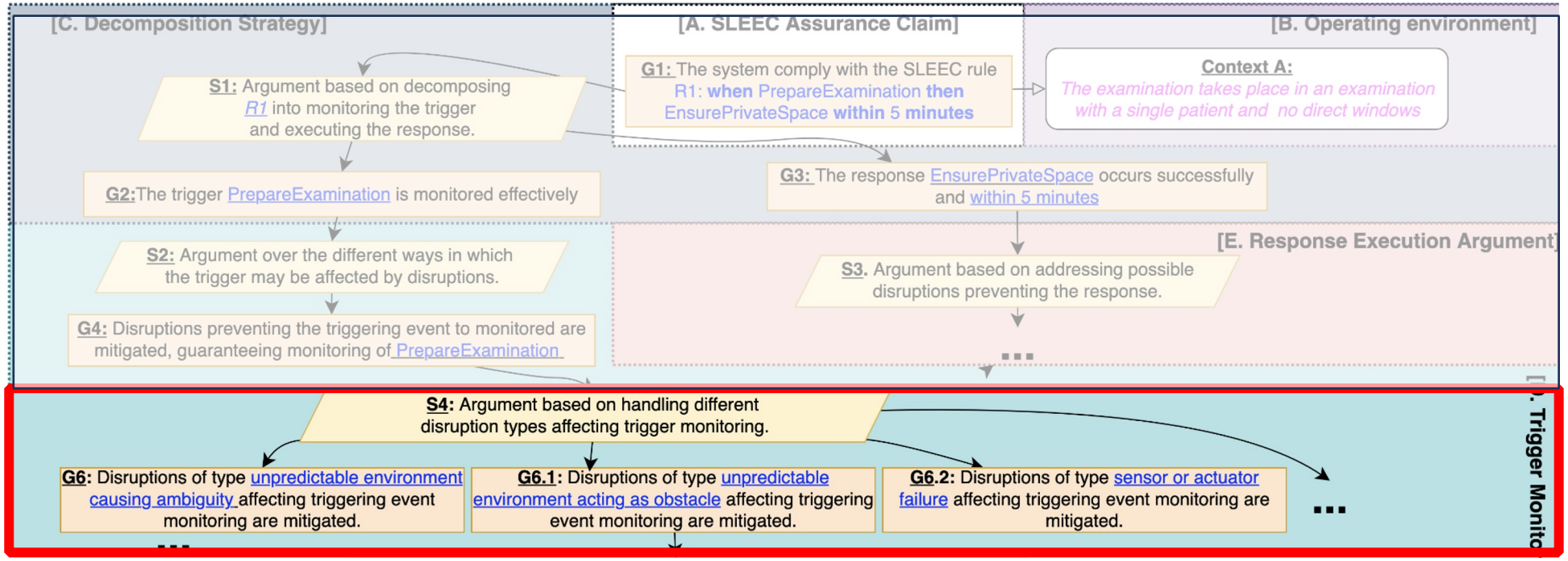
★ SLEEC rule: **when** *trigger* **then** *response*

★ Two possible ways to not comply with a SLEEC rule
- The system failed to monitor the trigger (*event* or *measures*)
- The system captured the response, but failed to perform the response (main response or defeaters)

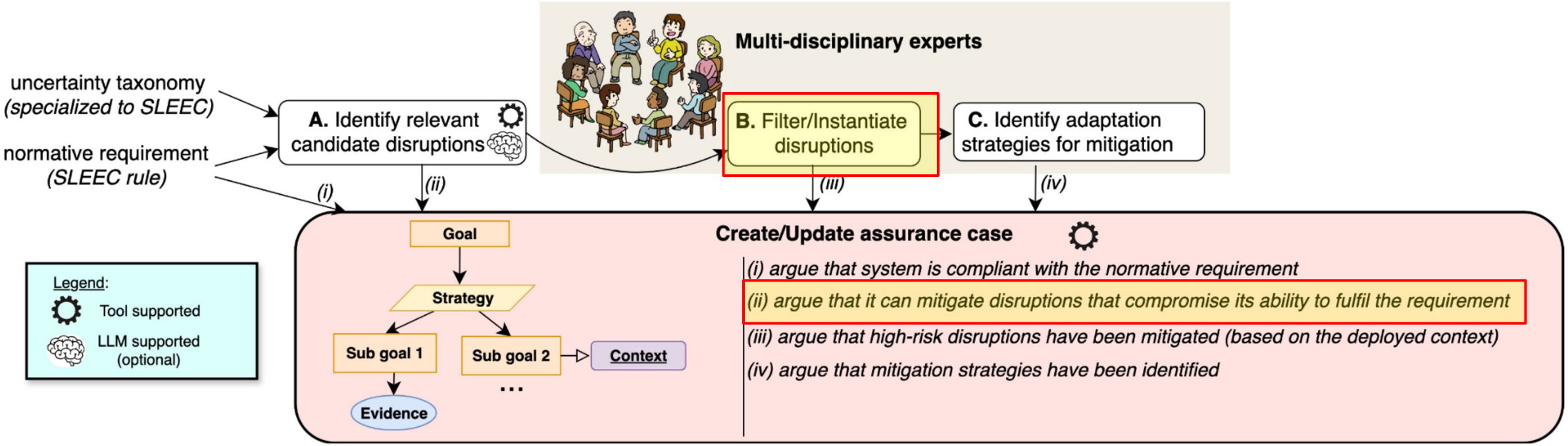
We defined 8 templates, to decompose the SLEEC rule into possible disruption classes, capturing when the trigger failed to be monitored (event, or measures), or the response failed (main response, or defeaters).

Identify relevant candidate disruptions





SLEEC-FIASCO: identify relevant disruptions



Disruption Template

Disruption template:	[UNCERTAINTY TYPE] affects the system capturing triggering [EVENT]
Disruption candidate *(instantiated with uncertainty class)*	Unexpected obstructive environment affects the system capturing PrepareExamination occurrence
E.g. Disruption 1	Sunlight glare in the room affects the system capturing PrepareExamination occurrence  L2
E.g. Disruption 2	Curtains closed affects the system capturing PrepareExamination occurrence  L1

LLM-based disruption recommender (**optional**)

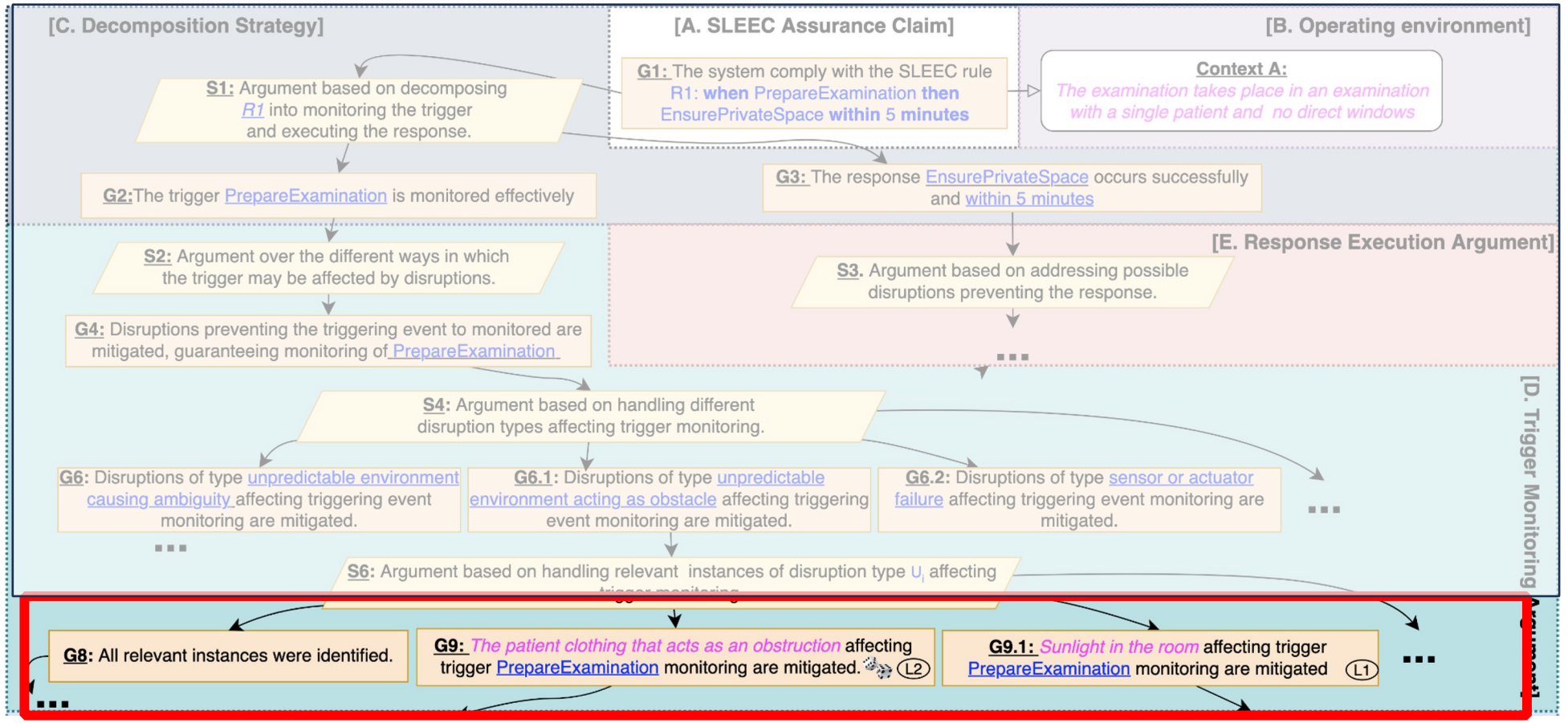
Use optionally our LLM-based disruption recommender for the identification of disruptions candidates.

- LLM component: mitigates "**blank page**" problem, **not replacing experts**
- Supports stakeholder **understanding of uncertainty types**
- Prompts based on ADVICE taxonomy, with curated examples from SEAMS exemplars
- Validation: tested on ALMI case study (manual review)

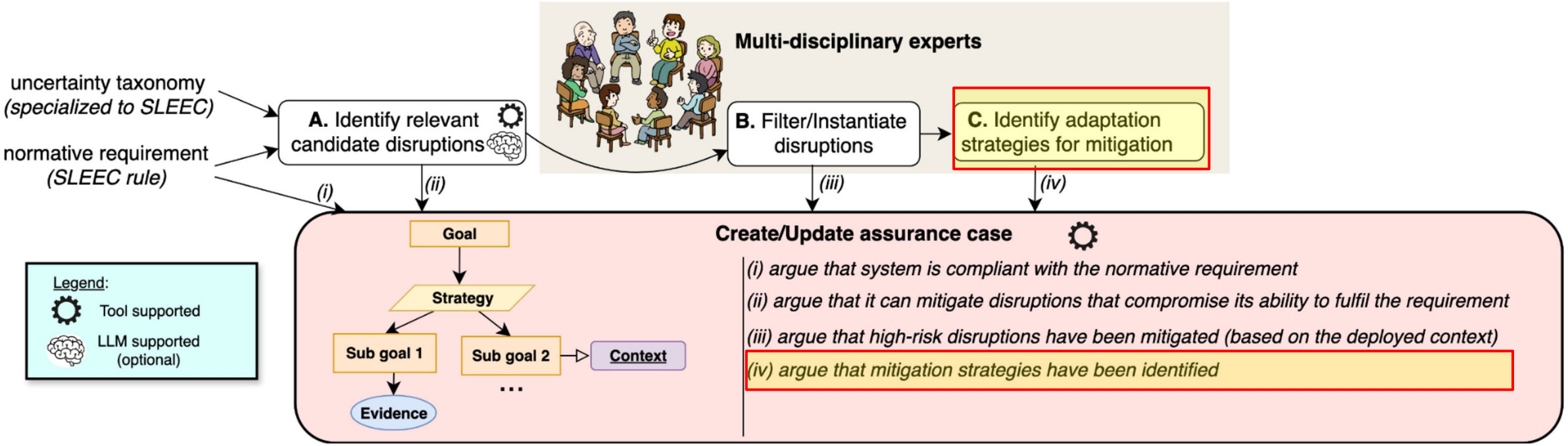


Filter/instantiate the disruption classes





SLEEC-FIASCO: identify relevant disruptions



Identify mitigation strategies

The advise uncertainty taxonomy can guide the selection of suitable mitigations.

- **Prevent / Avoid:** Identify and mitigate disruptions before deployment
- **Adapt / Reduce:** Use runtime strategies (fault tolerance, graceful degradation, proactive adaptation)
- **Accept / Delegate:** When unavoidable, accept risk or delegate (e.g., insurance, human intervention)

SLEEC requirement:
when GiveMedReminder **and** detectOtherPerson **then** AskConsent **within** 10 **minutes**

Another person is not detected due to poor lightning conditions (occlusion)
[Unexpected obstructive environment (Location); Aleatory (Nature); Full (Extent), L2]

Mitigations:

- PREVENT the disruption by using multiple sensors
- REDUCE the impact of the disruption by rechecking for person presence shortly after the initial scan, if a person is not detected and the environment conditions are ambiguous …

Mitigation strategy arguments



[C. Decomposition Strategy]

S1: Argument based on decomposing R1 into monitoring the trigger and executing the response.

G2: The trigger PrepareExamination is monitored effectively

S2: Argument over the different ways in which the trigger may be affected by disruptions.

G4: Disruptions preventing the triggering event to monitored are mitigated, guaranteeing monitoring of PrepareExamination

[A. SLEEC Assurance Claim]

G1: The system comply with the SLEEC rule R1: **when** PrepareExamination **then** EnsurePrivateSpace **within** 5 **minutes**

G3: The response EnsurePrivateSpace occurs successfully and within 5 minutes

[B. Operating environment]

Context A: *The examination takes place in an examination with a single patient and no direct windows*

[E. Response Execution Argument]

S3. Argument based on addressing possible disruptions preventing the response.

...

[D. Trigger Monitoring Argument]

S4: Argument based on handling different disruption types affecting trigger monitoring.

G6: Disruptions of type unpredictable environment causing ambiguity affecting triggering event monitoring are mitigated.

G6.1: Disruptions of type unpredictable environment acting as obstacle affecting triggering event monitoring are mitigated.

G6.2: Disruptions of type sensor or actuator failure affecting triggering event monitoring are mitigated.

...

S6: Argument based on handling relevant instances of disruption type U_i affecting trigger monitoring.

G8: All relevant instances were identified.

G9: *The patient clothing that acts as an obstruction* affecting trigger PrepareExamination monitoring are mitigated. L2

G9.1: *Sunlight in the room* affecting trigger PrepareExamination monitoring are mitigated L1

...

[F. Mitigation Argument]

S8: Argument based on decomposing disruption instance type into risk categories with corresponding mitigation approaches

S8.1: Argument based on decomposing disruption instance type into risk categories with corresponding mitigation approaches

G13: Mitigable disruption instance .

G12: Low-risk disruption instance.

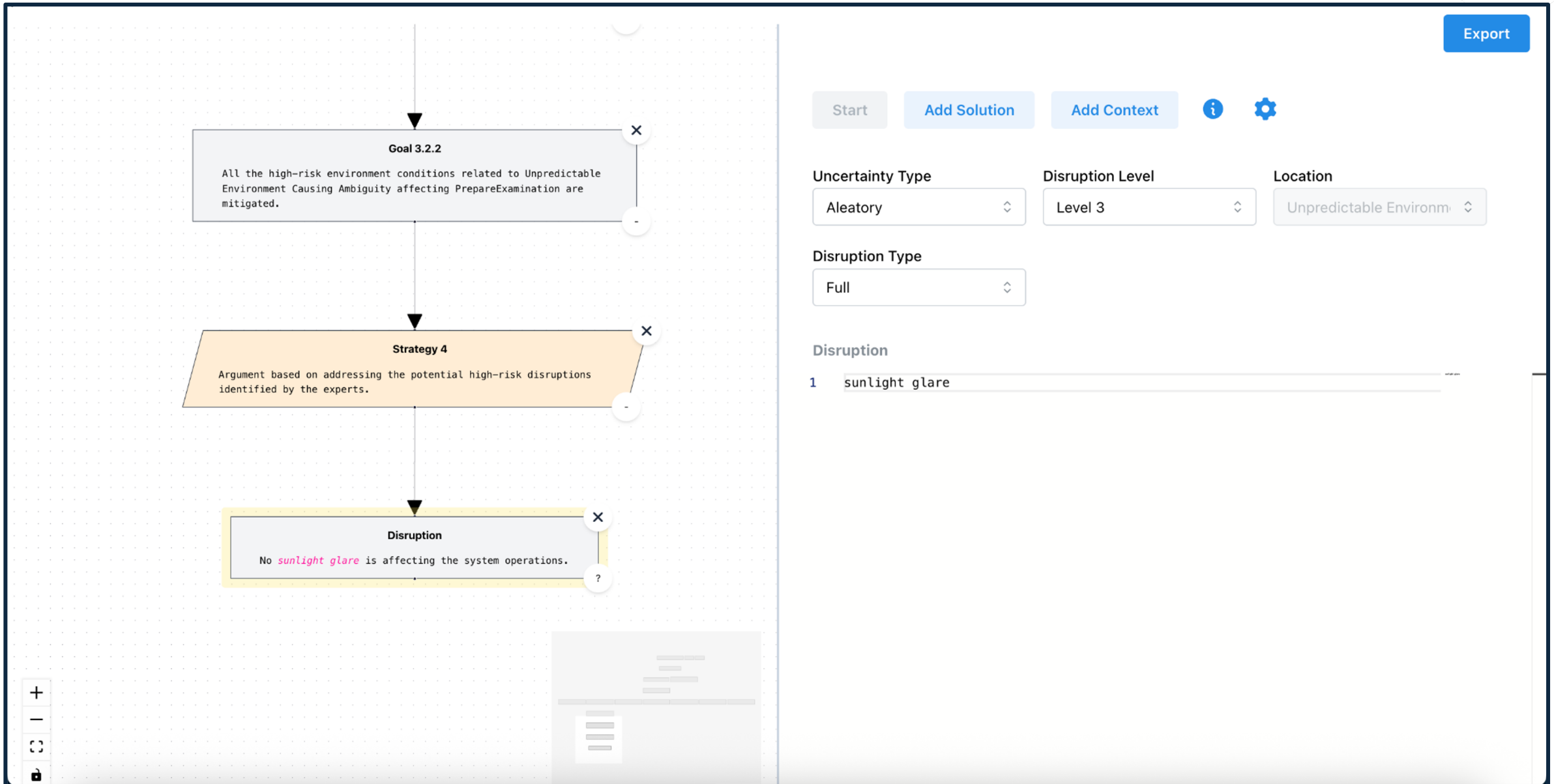
S9: Argument based on applicable mitigation strategies

e1: Low-risk disruption; accept and not mitigate.

G15: Mitigation strategy **prevent by** *combining face recognition with voice verification when the patient is not recognized* ensures compliance is preserved under disruption conditions.

G15.1: Mitigation strategy **avoid by** *calling for support when the patient is not recognized by the system* ensures compliance is preserved under disruption conditions.

SLEEC-FIASCO tool-set (with web interface)



Study (4 groups of multidisciplinary experts, 2 case studies)

24 hours

2 hours

experiments	group	cases	rule	disruption template†					uncertainty type‡							mitigation strategy*				
				T1	T2	T3	T4	T5	E1	E2	E3	SA1	SA2	SA3	SA4	M1	M2	M3	M4	M5
freeform-exp	(G1)	Casper	R1	1	3	-	1	-	1	3	0	1	1	1	0	4	2	3	4	2
freeform-exp	(G1)	Casper	R2	3	-	3	3	-	2	2	1	2	0	0	1	3	6	4	5	4
freeform-exp	(G1)	Casper	R3	1	1	-	3	-	0	2	0	1	1	1	1	1	3	3	5	1
freeform-exp	(G3)	Casper	R1	0	0	-	0	-	0	0	0	0	0	0	0	0	0	0	0	0
freeform-exp	(G3)	Casper	R2	0	-	2	0	-	2	0	0	0	0	0	0	1	1	0	0	2
freeform-exp	(G3)	Casper	R3	2	0	-	0	-	0	2	0	0	0	0	0	0	2	0	1	1
freeform-exp	(G2)	DAISY	R1	3	1	-	0	-	1	2	1	0	0	1	0	2	2	0	2	2
freeform-exp	(G2)	DAISY	R2	0	-	5	-	0	1	3	1	0	0	0	0	4	1	0	0	1
freeform-exp	(G2)	DAISY	R3	0	-	2	0	-	0	1	0	0	0	0	1	2	1	0	0	0
freeform-exp	(G4)	DAISY	R1	0	3	-	0	-	0	2	0	0	0	0	1	3	2	0	2	1
freeform-exp	(G4)	DAISY	R2	0	-	5	-	0	1	0	0	1	0	1	2	2	3	0	3	2
freeform-exp	(G4)	DAISY	R3	0	-	2	1	-	0	0	0	1	0	0	1	1	0	0	5	3
sleec-fiasco-exp	(G2)	Casper	R1	7	7	-	7	-	3	3	3	3	3	3	3	8	1	4	5	3
sleec-fiasco-exp	(G2)	Casper	R3	7	– time out –				1	1	1	1	1	1	1	4	0	2	1	1
sleec-fiasco-exp	(G2)	Casper	R2	– time out –																
sleec-fiasco-exp	(G4)	Casper	R1	8	7	-	5	-	3	3	3	3	3	2	3	13	2	0	11	4
sleec-fiasco-exp	(G4)	Casper	R3-R2	– time out –																
sleec-fiasco-exp	(G1)	DAISY	R1	14	7	-	7	-	7	6	3	3	3	3	3	9	1	5	7	6
sleec-fiasco-exp	(G1)	DAISY	R3	7	-	7	-	6	3	3	3	3	3	2	3	8	3	6	5	2
sleec-fiasco-exp	(G1)	DAISY	R2	– time out –																
sleec-fiasco-exp	(G3)	DAISY	R1	7	7	-	7	-	3	3	3	3	3	3	3	11	5	3	5	6
sleec-fiasco-exp	(G3)	DAISY	R3	7	-	7	7	-	3	3	3	3	3	3	3	12	2	5	6	4
sleec-fiasco-exp	(G3)	DAISY	R2	7	-	8	-	3	3	4	3	2	2	2	2	12	3	2	8	2

†number of disruptions identified per disruption template (T1--T5 as defined in Tbl. 3)
‡number of disruptions categorized by uncertainty type (E1: unpredictable environment acting as obstacle, E2: causing ambiguity, E3: totally unknown, SA1: sensor/actuator failure, SA2: imprecision, SA3: noise, SA4: inaccuracy)
*number of proposed mitigations per mitigation type (M1: Prevent, M2: Avoid, M3: Adapt, M4: Reduce, M5: Accept)

Results

SLEEC-relevant disruptions identification

- 45 SLEEC-relevant disruptions identified in freeform
  (22 DAISY, 23 Casper; 15 additional disruptions filtered as irrelevant)
- 156 SLEEC-relevant disruptions identified with SLEEC-FIASCO
  (48 Casper, 108 DAISY; all disruptions rule-relevant)
- Full coverage of disruption templates and ADVICE uncertainty types with the tool (for completed rules)
- 6 disruption templates left uncovered in freeform

Mitigation Identification

- 107 mitigations identified in freeform
  (102 classified into M1–M5; 5 new type: Defer)
- 197 mitigations identified with SLEEC-FIASCO
- Prevent and Reduce strategies most common in both settings
- Tool supported context-based justification of Accept mitigations

Takeaways

Verifying autonomous agent compliance with fixed SLEEC norms is already difficult – but not enough

Resilient agents may need to uphold different norms for each user, and must mitigate disruptions

We extended the SLEEC framework with

- a method and reusable software components for runtime agent adaptation (SLEEC-ADAPT)
- a tool-supported approach enabling multidisciplinary stakeholders to identify SLEEC-relevant disruptions and corresponding mitigations (avoiding a SLEEC-FIASCO)

Many remaining challenges

- normative agent adaptation to multi-user scenarios
- online disruption detection and mitigation
- extensions to foundation-model agents (agents without a predefined workflow or finite action set)